

BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE – PILANI, HYDERABAD
CAMPUS
EEE/INSTR F342: POWER ELECTRONICS, SEM-II, 2025-26

TUTORIAL- 2

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1-phase half wave and full wave full bridge diode rectifier

Q1. A single-phase full bridge diode rectifier supplies a very high inductive load such as a DC motor and has an armature current of 10Ω . The motor inductive load is so high that it draws a ripple free armature current. If it is supplied from 230V (RMS), 50 Hz source, Determine (a) average values of voltage and output current, (b) average and RMS values of diode currents (c) RMS values of output and input currents (d) Fourier expression of input current (derive) ϵ THD in % for the input current.

Q2. A single-phase full wave bridge rectifier is required to supply an average voltage of $V_{dc}=400\text{V}$ to a resistive load of $R=10\Omega$. Determine the voltage and current ratings of the diode and transformer (Transformer Utilisation Factor (TUF)).

Q3. The diodes in the single-phase full wave rectifier have a reverse recovery time of $50\mu\text{s}$ and the RMS input voltage of 120V. Determine the effect of the reverse recovery time on the average output voltage if the supply frequency is (a) 60Hz and (b) 2 kHz.

Q4. A single-phase full bridge diode rectifier circuit is fed by a 220V, 50 Hz supply. The load has 10Ω resistance. (a) Determine the RMS, peak and PIV for the diodes under ideal condition. (b) If the forward voltage drop for each diode is 0.8V, determine the value of average output voltage? (c) Assuming the factor of safety to be 1.2 calculate both (a) and (b) cases.